

Understanding Cybersecurity Concepts



Learning Objectives

By the end of this lesson you will...

Describe the **AAA framework** and provide examples.

Explain the **CIA triad** and its importance.

Discuss **password hygiene** and share good practices.



Key Vocabulary

AAA. A framework that includes Authentication, Authorization, and Accounting, which helps manage user access.

CIA. A model that stands for Confidentiality, Integrity, and Availability, essential for data security.

Authentication. The process of verifying the identity of a user or device.

Authorization. The determination of what an authenticated user is allowed to access.

Accounting. The tracking of user activities and resource usage.

Confidentiality. Ensuring that sensitive information is not disclosed to unauthorized individuals.

Integrity. Maintaining the accuracy and reliability of data over time.

Availability. Ensuring that information is accessible to authorized users when needed.

Password Hygiene. The practice of maintaining strong and secure passwords to protect personal information.



Poll

How confident are you in your password hygiene?

- ☐ Very confident!
- ☐ Somewhat confident.
- ☐ Not very confident.
- ☐ I need to improve!

Understanding the AAA Framework

AAA stands for **Authentication**, **Authorization**, and **Accounting**. These are key concepts in managing security.



Authentication

Authentication is all about verifying who you are. For example, when you log into your email, you enter a username and password to prove your identity.



Authorization

Authorization is a crucial aspect of security that determines what actions you can perform after logging in. It ensures that users have the appropriate permissions to access certain features or settings.

You can read messages in your email after logging in.

You may not be able to change account settings without permission.

Authorization helps protect sensitive information by restricting access.



Accounting

Accounting tracks what you do while logged in. This includes logging your actions, like sending emails or changing settings, so there's a record of your activities.



Real-World Example of AAA

In the context of banking, we can see how the concepts of Authentication, Authorization, and Accounting work together. Here's a breakdown of each term:

Authentication: This is your ID and password used to log in to your bank account.

Authorization: This determines what you can access, such as checking your balance.

Accounting: This is the record of your transactions.



AAA Framework Discussion

Share your thoughts on how the AAA framework applies to your online activities. How do you see it in action?

^ Instructions



Collaborate Board

AAA Framework Discussion

Multiple Choice

What is the primary purpose of authentication in cybersecurity?

- ☐ To encrypt data during transmission.
- ☐ To manage user permissions.
- ☐ To verify the identity of users or systems.
- ☐ To monitor user activities.

Which of the following best describes authorization?

- ☐ It verifies the identity of users.
- ☐ It determines access rights and permissions.
- ☐ It logs user activities for auditing.
- ☐ It encrypts sensitive information.

What does accounting refer to in the context of cybersecurity?

- ☐ Verifying user identities.
- ☐ Granting access to resources.
- ☐ Tracking user activities and resource usage.
- ☐ Encrypting data for security.

True or False: Authentication and authorization are the same processes in cybersecurity.

- ☐ True
- ☐ False

What is the relationship between authentication, authorization, and accounting?

- ☐ They are unrelated processes.
- ☐ They work together to secure systems.

True or False: Authorization occurs before authentication in the security process.

☐ False

☐ True

Understanding the CIA Triad

The **CIA Triad** represents three key principles: **Confidentiality**, **Integrity**, and **Availability**. These principles are vital for **protecting information** in various contexts, ensuring that data is kept secure, accurate, and accessible when needed.



Confidentiality

Confidentiality means keeping information secret from those who shouldn't see it. For example, your **medical records** should only be accessible to you and your **doctor**.



Integrity

Integrity ensures that data is **accurate** and **trustworthy**. For instance, when you update your address in a system, it should reflect the **correct information** without errors.



Availability

Availability means that information is accessible when you need it. For example, you should be able to access your online classes anytime.



Real-World Example of CIA

In banking, it is crucial to ensure that systems uphold three key principles: confidentiality, integrity, and availability. These principles help protect sensitive information and maintain trust in the banking system.

Confidentiality: Keeping account details private and secure.

Integrity: Ensuring accurate balances and preventing unauthorized changes.

Availability: Guaranteeing access to accounts at all times.



Multiple Choice

What does confidentiality refer to in information security?

- ☐ Making data public for transparency
- ☐ Maintaining data accuracy over time
- ☐ Protection of sensitive information from unauthorized access
- ☐ Ensuring data is available to everyone

Which of the following best describes integrity in the context of information security?

- ☐ Making data available to all users
- ☐ Maintaining the accuracy and consistency of data
- ☐ Ensuring data is accessible at all times
- ☐ Protecting data from being viewed by unauthorized users

What is the primary focus of availability in information security?

- ☐ Ensuring authorized users can access information when needed
- ☐ Preventing unauthorized access to sensitive data
- ☐ Maintaining the confidentiality of data
- ☐ Making data available to everyone without restrictions

True or False: Integrity means that data can be modified by anyone at any time.

- ☐ True
- ☐ False

Importance of Password Hygiene

Good password hygiene is crucial for protecting your personal and sensitive information from unauthorized access.



Examples of Good Practices

To maintain strong security, it is essential to follow certain good practices regarding passwords. These practices help protect your personal information from unauthorized access.

Use **complex passwords** that include letters, numbers, and symbols.
Change your **passwords regularly** to enhance security.
Never **share your passwords** with others.



Consequences of Poor Password Hygiene

Maintaining strong passwords is crucial for protecting your personal information. Weak passwords can lead to serious consequences, including:

- Unauthorized access to accounts

- Identity theft, where someone pretends to be you

- Data breaches that expose sensitive information

- Increased risk of financial loss



Real-World Example of Password Breach

A recent data breach highlighted the dangers of using **weak passwords**. In this incident, hackers exploited easily guessable passwords, leading to unauthorized access to sensitive information. This breach affected many users and raised concerns about online security.

To prevent such breaches, it is crucial to use **strong passwords** that combine letters, numbers, and symbols. Additionally, implementing **two-factor authentication** can add an extra layer of security, making it harder for hackers to gain access.



Multiple Choice

What is password hygiene?

- ☐ The act of sharing passwords with friends.
- ☐ Using the same password for all accounts.
- ☐ The practice of keeping passwords secure and private.
- ☐ Changing passwords every month without reason.

Which of the following is a good practice for creating a strong password?

- ☐ Using the same password for multiple accounts.
- ☐ Using your name and birthdate.
- ☐ Creating a password that is easy to remember.
- ☐ Using a combination of letters, numbers, and symbols.

True or False: It is safe to write down your passwords and keep them in your wallet.

☐ True

☐ False

What should you do if you suspect your password has been compromised?

- ☐ Ignore it and hope for the best.
- ☐ Change the password immediately and monitor your account.

Which of the following is NOT a recommended practice for password hygiene?

- ☐ Changing passwords regularly.
- ☐ Using the same password for different accounts.
- ☐ Using a password manager.
- ☐ Creating unique passwords for each account.

What does the 'C' in CIA stand for in cybersecurity?

- ☐ Control
- ☐ Compliance
- ☐ Communication
- ☐ Confidentiality

Which of the following best describes Integrity in the context of the CIA triad?

- ☐ Keeping data secret from unauthorized users
- ☐ Backing up data regularly
- ☐ Ensuring data is accurate and reliable
- ☐ Making data available to users

Open Ended Question

What are your personal strategies for maintaining password hygiene? Share your thoughts!

Please enter your answer here.



Time To Climb

Lesson Summary and Call to Action

In today's lesson, we explored key concepts related to online safety. Here are the main topics we covered:

The **AAA framework** which stands for Authentication, Authorization, and Accounting.

The **CIA triad** focusing on Confidentiality, Integrity, and Availability.

The importance of **password hygiene** for protecting personal information.



Multiple Choice

What is password hygiene?

- ☐ Practices to ensure password security
- ☐ Writing passwords down in an obvious place
- ☐ Sharing passwords with friends
- ☐ Using the same password for all accounts

What does authentication refer to in cybersecurity?

- ☐ Verifying the identity of a user
- ☐ Monitoring user activity
- ☐ Granting access to all users
- ☐ Encrypting data

What is the purpose of authorization in a security context?

- ☐ Creating user accounts
- ☐ Monitoring network traffic
- ☐ Determining user permissions
- ☐ Encrypting sensitive information

What does accounting mean in the context of cybersecurity?

- ☐ Verifying user identities
- ☐ Tracking user activities
- ☐ Creating strong passwords
- ☐ Granting access to resources

What are the three components of the CIA triad?

- ☐ Encryption, Decryption, Access Control
- ☐ Confidentiality, Integrity, Availability
- ☐ Authentication, Authorization, Accounting
- ☐ Monitoring, Auditing, Reporting

What does integrity mean in the context of the CIA triad?

- ☐ Encrypting data for security
- ☐ Making data accessible to everyone
- ☐ Sharing data openly
- ☐ Accuracy and trustworthiness of data

Why is it important to regularly update passwords?

- ☐ To share with more people
- ☐ To avoid using strong passwords
- ☐ To reduce the risk of unauthorized access
- ☐ To make it easier to remember passwords